Chapter 11. Coloring and Matching

Finding a Good Match

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11.1 Introduction

Motivation. Assigning frequencies to towers so that nearby towers differ is colouring the vertices of a conflict graph. The least number of colours needed, the chromatic number, measures how constrained the graph is.

A phone company assigns one of four frequencies to each of ten towers; towers within 50 miles must use different frequencies. Join two towers by an edge if they are within 50 miles, and colour each vertex by its frequency.

Proposition 11.1

The four frequencies can be assigned to the ten towers iff the vertices of the graph can be coloured with four colours so that no two adjacent vertices have the same colour.

Proof. A frequency assignment is a colouring with four colours, and the constraint "towers within 50 miles differ" is exactly "adjacent vertices differ". ■

Definition 11.2 (Chromatic number)

The **chromatic number** $\chi(H)$ is the least k such that the vertices of H can be coloured with k colours with adjacent vertices coloured differently (a **proper colouring**). H is **k-colourable** if $\chi(H) \leq k$.

Example 11.3

The pentagon C_5 has $\chi = 3$. Two colours would have to alternate around a cycle of odd length, which is impossible; three colours suffice (red, blue, red, blue, green).

Components can be coloured independently and multiple edges add no constraints, so we may assume graphs are connected and simple.

11.2 Bipartite graphs

Motivation. Two colours suffice exactly when there is no odd cycle. Bipartite graphs can be dense (up to $n^2/4$ edges), but one edge more forces not just an odd cycle but a triangle, and in fact m triangles at once.

Definition 11.4 (Bipartite)

A 2-colourable graph is **bipartite**: its vertex set splits into **colour classes** A, B such that every edge has one end in A and one in B.

Trees are bipartite (colour by parity of the distance from a fixed vertex), as are even cycles. Triangles and all odd cycles are not.

Theorem 11.5 (König)

A graph is bipartite iff it contains no cycle of odd length.

Proof.

Step 1 (only if). Colours must alternate along a cycle $A_1A_2\cdots A_{2m+1}$, so A_1 and A_{2m+1} get the same colour although they are adjacent.

Step 2 (colouring by distance). Let G (wlog connected) have no odd cycle. Fix V, and colour W red if its distance $d(V,W)$ is even, blue if odd. Suppose an edge PQ joins two vertices of the same colour. Shortest paths $V \to P$ and $V \to Q$, followed by the edge PQ, form a closed walk of odd length (two lengths of equal parity plus one).

Step 3 (odd closed walks contain odd cycles) [Supplement]. A closed walk of odd length contains an odd cycle. Induction on its length: if no vertex repeats (other than start = end), it is a cycle. Otherwise split it at a repeated vertex into two shorter closed walks whose lengths add up to an odd number; one of them is odd. This contradicts the assumption, so the colouring is proper. ■

Theorem 11.6

A simple bipartite graph on n vertices has at most $\frac{n^2}{4}$ edges if n is even and at most $\frac{n^2-1}{4}$ if n is odd.

Proof. Take an extremal G with colour classes of sizes a and $n-a$. Every edge between the classes is present (otherwise add it), so G has $a(n-a)$ edges. This quadratic in a is maximised at $a = \lfloor n/2 \rfloor$, giving $\frac{n^2}{4}$ or $\frac{n^2-1}{4}$. ■

The extremal graphs are the **complete bipartite graphs** $K_{a,b}$. A graph on $2m$ vertices with m^2 edges can be bipartite ($K_{m,m}$); with more edges it has an odd cycle, and in fact more.

Lemma 11.7 (Mantel)

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains a triangle.

Proof.

Step 1 ($m = 2$). $H \subseteq K_4$ has at least 5 edges, so it is not bipartite (Theorem 11.6) and has an odd cycle, which on 4 vertices must be a triangle.

Step 2 (induction). Take an edge FG. If $\deg F + \deg G > 2m$, then F and G have a common neighbour T (they have at least $2m-1$ neighbours in the other $2m-2$ vertices, counting G and F themselves), and FGT is a triangle. Otherwise deleting F and G removes at most $2m-1$ edges (FG is counted twice), leaving $2m-2$ vertices and at least $m^2+1-(2m-1) = (m-1)^2+1$ edges. By induction there is a triangle. ■

Theorem 11.8

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains at least m triangles.

Proof. Assume exactly m^2+1 edges (deleting edges destroys no triangle count in our favour) and use induction on m; for $m = 2$, $H = K_4$ minus an edge, which has 2 triangles. By Lemma 11.7 there is a triangle ABC; we need $m-1$ more. Let the number of edges between ABC and the other $2m-3$ vertices be $2m-3+x$. Each outside vertex sending two edges to ABC spans a triangle with them, and by pigeon-holing at least x outside vertices do so: at least x triangles of this kind. Let R be the graph on the outside vertices.

Step 1 ($x \geq m-1$). Done.

Step 2 ($1 \leq x < m-1$). At most $3m-5$ edges meet ABC from outside, so R has at least $m^2+1-(3m-5)-3 = (m-1)(m-2)+1$ edges on $2m-3$ vertices. Deleting a vertex of minimum degree in R removes at most the average degree. So the remaining R' on $2m-4$ vertices has more

than $(m-1)(m-2)\frac{2m-4}{2m-3} = (m-2)^2\frac{2m-2}{2m-3} > (m-2)^2$ edges. By induction R' has $m-2$ triangles; with the $x \geq 1$ triangles above we have $m-1$.

Step 3 ($x \leq 0$). At most $2m-3$ edges join ABC to R, so R has at least $m^2+1-(2m-3)-3 = (m-1)^2$ edges. If no such edge exists, R has $m^2-2 \geq (m-1)^2+1$ edges and R plus any vertex of ABC is a graph on $2m-2$ vertices to which induction applies. Otherwise add to R a vertex of ABC adjacent to R, getting $2m-2$ vertices and at least $(m-1)^2+1$ edges, hence $m-1$ further triangles by induction. ∎

So as soon as one triangle is guaranteed, m of them are.

11.3 Matchings in bipartite graphs

Motivation. Filling job openings with qualified applicants means choosing disjoint edges in a bipartite graph. Hall's condition (every group of openings has at least as many qualified applicants) is obviously necessary, and remarkably also sufficient. Augmenting paths certify maximality, and deferred acceptance produces stable assignments.

With jobs and applicants as the two classes and edges for qualifications, a set of filled jobs is a set of vertex-disjoint edges.

Proposition 11.9

All m openings can be filled simultaneously iff the job–applicant graph has m vertex-disjoint edges.

Proof. A hiring fills each job with one qualified person and gives each person at most one job. These are exactly the edges "job–person hired for it", and they are pairwise vertex-disjoint; conversely, m such edges are a complete hiring. ∎

Definition 11.10 (Matching)

A set S of edges of G, no two sharing a vertex, is a **matching** (independent set of edges). It is **perfect** if it covers every vertex.

Definition 11.11 (Matching X into Y)

For a bipartite $G = (X, Y)$, a matching covering all of X is a **perfect matching of X into Y**; we say X **can be matched into Y**.

11.3.1 Bipartite graphs with perfect matchings

For $T \subseteq X$ let $N(T) = N_G(T)$ be the set of neighbours of T.

Proposition 11.12

If X can be matched into Y, then $|T| \leq |N(T)|$ for all $T \subseteq X$.

Proof. A matching of X into Y matches T injectively into $N(T)$. ∎

Theorem 11.13 (Hall's theorem)

X can be matched into Y iff $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof (Halmos–Vaughan). "Only if" is Proposition 11.12. For "if", induct on $|X|$; $|X| \le 1$ is trivial.

Step 1 (strict surplus). Suppose $|T| < |N(T)|$ for all nonempty proper $T \subset X$. Take an edge xy, $x \in X$, and $G' = G - x - y$. For nonempty $A \subseteq X - x$, $|N_{G'}(A)| \ge |N_G(A)| - 1 \ge |A|$. By induction $X - x$ can be matched into $Y - y$ in G'; add xy.

Step 2 (a tight set). Otherwise some nonempty proper $B \subset X$ has $|B| = |N(B)|$. Let G_1 be induced by $B \cup N(B)$ and G_2 the graph obtained by deleting $B \cup N(B)$. In G_1, $N_{G_1}(T) = N_G(T)$ for $T \subseteq B$, so Hall's condition holds. In G_2, for $U \subseteq X - B$, $N_G(U \cup B)$ is the disjoint union of $N_{G_2}(U)$ and $N_G(B)$, so

$$|N_{G_2}(U)| = |N_G(U \cup B)| - |N_G(B)| \ge |U| + |B| - |B| = |U|.$$

By induction B is matched onto $N(B)$ and $X - B$ into $Y - N(B)$; together they match X into Y. ■

A matching is **maximal** if no edge can be added to it, and **maximum** if no matching is larger. Every maximum matching is maximal, but not conversely: a path with three edges has the maximal matching consisting of its middle edge, whereas the two end edges form a larger matching.

Definition 11.14 (Alternating path)

For a matching M, a path $v_1 v_2 \cdots v_r$ is **M-alternating** if its edges alternate between M and not M.

Definition 11.15 (Augmenting path)

An M-alternating path is **M-augmenting** if it starts and ends at vertices not covered by M.

Swapping the edges of an augmenting path P ($M \mapsto M \triangle P$) gives a matching with one more edge.



Figure. *An M-augmenting path (bold: M, dashed: not in M) between two uncovered vertices. Swapping bold and dashed edges increases the matching by one.*

Theorem 11.16 (Berge)

A matching M in a simple graph is maximum iff there is no M-augmenting path.

Proof. "Only if": an augmenting path enlarges M. "If": let M' be a maximum matching and consider $M \triangle M'$. Every vertex meets at most one edge of each, so the components are paths and cycles alternating between M and M', and the cycles are even. A path with more M'-edges than M-edges would be M-augmenting, and one with more M-edges would be M'-augmenting, contradicting maximality of M'. So every component has equally many edges of M and M', and $|M| = |M'|$. ■

11.3.2 Stable matchings in bipartite graphs

n candidates and n jobs; each candidate ranks all jobs and each hiring manager ranks all candidates.

Definition 11.17 (Stable matching)

In a complete bipartite $G(X,Y)$ where every vertex ranks the other side, a perfect matching M is **stable** if there are no $x \in X$, $y' \in Y$ such that (a) $xy \in M$ but x prefers y' to y, and (b) $x'y' \in M$ but y' prefers x to x' (no pair would both rather be together).

Gale–Shapley (deferred acceptance). Each unmatched candidate applies to the best job not yet tried; each manager keeps the best applicant so far (tentatively) and rejects the others. The process ends with every candidate placed: an unplaced candidate C always has a job it has not yet applied to.

The result is stable. Suppose $xy, x'y' \in M$, x prefers y' to y, and y' prefers x to x'. Then x applied to y' before y. If y' accepted x then, it later dropped x for someone it prefers, and its final hire x' is at least as good, hence better than x, a contradiction. If y' rejected x, it already held someone better than x, and its final hire is at least as good, again a contradiction. With unequal sides, add fictitious jobs or candidates ranked last by everyone. In non-bipartite graphs, stable matchings need not exist.

Remark

Remark[Algorithmic note] Gale–Shapley runs in $O(n^2)$ applications, since each candidate applies to each job at most once.

11.4 More than two colours

Motivation. Which k-colourable graphs on n vertices have the most edges? The balanced complete k-partite graph, and any graph with more edges contains K_{k+1} (Turán). Large chromatic number does not, however, require a large clique.

Write $n = tk + r$ with $0 \le r \le k-1$ and split n vertices into k classes, r of size $t+1$ and $k-r$ of size t. Join vertices iff they lie in different classes; this is the **complete k-partite (Turán) graph** H, with

$$T(n,k) = \frac{k-1}{2k}\,n^2 - \frac{r(k-r)}{2k} \tag{11.1}$$

edges. [Supplement] *Reason:* H has $\frac{1}{2}(n^2 - \sum(\text{class sizes})^2) = \frac{1}{2}(n^2 - r(t+1)^2 - (k-r)t^2)$ edges, and substituting $n = tk + r$ gives (11.1). H is k-colourable.

Theorem 11.18 (Turán)

A simple graph on n vertices with more than $T(n,k)$ edges contains K_{k+1}; in particular it is not k-colourable.

Proof.

Step 1. Let G be K_{k+1}-free with n vertices and the maximum number of edges; we show $e(G) \le T(n,k)$ by induction on $t = \lfloor n/k \rfloor$. For $t = 0$, $n < k$ and $T(n,k) = \binom{n}{2}$.

Step 2. For $t \ge 1$, G contains a K_k, call it S. Indeed, if G is complete then $n \ge k$ gives one; otherwise adding a missing edge creates a K_{k+1}, whose other $k-1$ vertices together with one endpoint of that edge form a K_k in G.

Step 3. Edges within S: $\binom{k}{2}$. Each vertex outside S has at most $k-1$ neighbours in S (else K_{k+1}). $G - S$ is K_{k+1}-free on $n-k = (t-1)k + r$ vertices, so by induction it has at most $T(n-k,k)$ edges. Hence

$$e(G) \le \binom{k}{2} + (n-k)(k-1) + T(n-k,k) = T(n,k). \tag{11.2}$$

Step 4 (the identity (11.2)) [Supplement]. Remove one vertex from each class of the Turán graph H on n vertices. What remains is the Turán graph on $n-k$ vertices (class sizes t and $t-1$, same r). The removed edges are the $\binom{k}{2}$ among the removed vertices plus $k-1$ for each of the $n-k$ remaining vertices (one to every removed vertex outside its own class). ■

Odd cycles of length ≥ 5 have $\chi = 3$ without containing K_3. In a sense, odd cycles and complete graphs are the only obstacles:

Theorem 11.19 (Brooks)

If G is connected, neither a complete graph nor an odd cycle, and all degrees are at most d with $d \geq 3$, then $\chi(G) \leq d$.

Proof [Supplement] (Lovász). The book leaves this to a guided exercise.

Step 1 (greedy towards a root). If some vertex w has degree $< d$, order the vertices by non-increasing distance from w and colour greedily with colours $[d]$. Every vertex other than w has a neighbour closer to w, coloured later, so at colouring time at most $d-1$ of its neighbours are coloured; w itself has fewer than d neighbours. So d colours suffice. Hence assume G is d-regular.

Step 2 (a cut vertex). If v is a cut vertex, let $C_1, \ldots, C_m$ be the components of $G - v$ and $G_i = G[C_i \cup \{v\}]$. In G_i the vertex v has degree $< d$, so G_i is d-colourable by Step 1 (or is a complete graph on at most d vertices). Permute colours so that v gets the same colour in all G_i and combine.

Step 3 (the key triple). Let G be 2-connected. We find w with non-adjacent neighbours u, v such that $G - u - v$ is connected.

If G is 3-connected: G is connected and not complete, so some w has two non-adjacent neighbours u, v (on a shortest path between non-adjacent vertices), and $G - u - v$ is connected.

If G has a 2-vertex cut $\{w, y\}$: $G - w$ is connected with cut vertex y, so its block structure has at least two leaf blocks. Each leaf block contains a vertex adjacent to w that is not a cut vertex of $G - w$, since otherwise G would have a cut vertex. Choose such u, v in two different leaf blocks; they are non-adjacent. Every vertex has degree $\geq d - 1 \geq 2$ in $G - w$, so leaf blocks are not single edges, and deleting u, v keeps $G - w$ connected. Finally w has $d \geq 3$ neighbours, so it keeps a neighbour in $G - w - u - v$, and $G - u - v$ is connected.

Step 4 (colouring). Colour u, v with colour 1. Order the rest of $G - u - v$ by non-increasing distance from w (in $G - u - v$), ending with w, and colour greedily from $[d]$. Each vertex $\neq w$ has a later neighbour, so at most $d - 1$ coloured neighbours. All d neighbours of w are coloured, but u, v share a colour, so they use at most $d - 1$ colours. Hence d colours suffice. ■

For every n there are triangle-free graphs with chromatic number n, so high chromatic number forces a high degree but not a large clique.

11.5 Matchings in graphs that are not bipartite

Motivation. For general graphs Hall's condition is replaced by Tutte's: deleting any s vertices may leave at most s odd components. Necessity is a parity count; for sufficiency one studies the maximal graphs without a perfect matching, whose structure turns out to be completely rigid.

For a vertex set S, $G - S$ deletes S and its edges, and $c_o(G - S)$ is the number of components of $G - S$ with an odd number of vertices.

Theorem 11.20 (Tutte)

G has a perfect matching iff $c_o(G-S) \le |S|$ for every vertex set S.

Only if. In a perfect matching, every odd component of $G-S$ has a vertex matched outside the component, i.e. into S, and these partners in S are distinct.

A graph is **saturated non-factorizable** if it has no perfect matching but adding any new edge creates one.

Lemma 11.21

If G is saturated non-factorizable and S is the set of vertices adjacent to all other vertices, then the components of $G-S$ are complete graphs.

Proof. Let ab, bc be edges of $G-S$ and suppose $ac \notin E$. Since $b \notin S$, some d is not adjacent to b. $G+ac$ has a perfect matching $F_1 \ni ac$ and $G+bd$ has one, $F_2 \ni bd$. $F_1 \triangle F_2$ is a union of even alternating cycles; let $C_1 \ni ac$ and $C_2 \ni bd$.

Step (a) $C_1 \neq C_2$. $F_1 \triangle C_1$ is a perfect matching avoiding ac and bd, i.e. one of G, a contradiction.

Step (b) $C_1 = C_2$. Traverse C_1 from b through d until reaching a or c, say a, along a path P. Then $P \cup ab$ is F_2-alternating and closed, and $F_2 \triangle (P \cup ab)$ is a perfect matching of G avoiding bd (it uses $ab \in E$), again a contradiction. ■

Theorem 11.22

G is saturated non-factorizable iff either (a) G is complete with an odd number of vertices, or (b) G has an even number of vertices and consists of vertex-disjoint complete graphs $S_0, G_1, \dots, G_k$ with $k = |S_0| + 2$, each $|G_i|$ odd, and every vertex of every G_i joined to every vertex of S_0.

Proof.

Step 1 (odd order). No graph with an odd number of vertices has a perfect matching, and only K_n admits no new edge.

Step 2 (even order: structure). Let $S_0 = S$ as in Lemma 11.21; then S_0 is complete, the components G_i of $G-S$ are complete, and S_0 is joined to everything. If $c_o(G-S) \le |S|$, a perfect matching would exist: match one vertex of each odd G_i to distinct vertices of S, the rest of each complete G_i internally, and the leftover vertices of S (an even number, by parity) among themselves. Hence $c_o > |S|$, so by parity $c_o \ge |S| + 2$. If $c_o > |S| + 2$, or if there were an even component, adding an edge between two components would leave $c_o(G'-S) > |S|$, so G' would still have no perfect matching, contradicting saturation. So $k = |S_0| + 2$, all odd.

Step 3 (conversely) [Supplement]. A graph as in (b) has $c_o(G-S_0) = |S_0| + 2 > |S_0|$, so it has no perfect matching. Missing edges only join different G_i, G_j. Adding $g_i g_j$ and using it, the remaining $|S_0|$ odd components each match one vertex to a distinct vertex of S_0, and everything else matches inside the complete pieces: a perfect matching. So G is saturated. ■

Proof of Theorem 11.20 ("if", Hetyei–Lovász). Suppose G satisfies Tutte's condition but has no perfect matching. $S = \emptyset$ shows that $|V(G)|$ is even. Add edges as long as no perfect matching arises, reaching a saturated non-factorizable $G' \supseteq G$ on the same vertices, as in Theorem 11.22(b). Let S' be its set of universal vertices. $G'-S'$ has $|S'|+2$ odd complete components. Removing the added edges may split these components, but every odd component of $G'-S'$ contains an odd component of $G-S'$. Hence $c_o(G-S') \ge |S'|+2 > |S'|$, contradicting Tutte's condition. ■

11.6 Selected exercises

Motivation. Hall's condition is rarely checked set by set; in the two book exercises below it follows from a single double count of edges, and the second one turns this into a decomposition theorem for matrices.

Exercise 11.4 (A degree condition for Hall)

Let G be a bipartite graph with colour classes X, Y in which every vertex of X has degree at least as large as every vertex of Y, and every vertex of X has at least one neighbour. Prove that X can be matched into Y.

Solution. Let $a = \min_{x \in X} \deg x$ and $b = \max_{y \in Y} \deg y$; by assumption $a \geq b$ and $a \geq 1$. We verify Hall's condition (Theorem 11.13).

Step 1 (double count). Let $T \subseteq X$ and let E_T be the set of edges with an end in T. Each vertex of T has at least a edges, and an edge has only one end in X, so $|E_T| \geq a|T|$. The other ends of these edges lie in $N(T)$, and each vertex of $N(T)$ is the end of at most b of them, so $|E_T| \leq b\,|N(T)|$.

Step 2 (Hall's condition). Hence $a|T| \leq b|N(T)| \leq a|N(T)|$, and dividing by $a \geq 1$ gives $|T| \leq |N(T)|$. By Theorem 11.13, X can be matched into Y.

The extra assumption is needed only to exclude the trivial case without edges, where $a = b = 0$. ■

Exercise 11.9 (Magic squares are sums of permutation matrices)

Let A be an $n \times n$ matrix with nonnegative integer entries in which every row and every column has the same sum r. Prove that A is a sum of r permutation matrices.

Solution. Induction on r. For $r = 0$, $A = 0$ is the empty sum. Let $r \geq 1$.

Step 1 (a bipartite graph). Let G be the bipartite graph with colour classes $X = \{x_1, \dots, x_n\}$ (rows) and $Y = \{y_1, \dots, y_n\}$ (columns), with $x_i y_j$ an edge iff $a_{ij} > 0$.

Step 2 (Hall's condition). Let $T \subseteq X$. The entries in the rows of T add up to $r|T|$. Every nonzero such entry lies in a column of $N(T)$, and the entries of these columns add up to $r|N(T)|$. All entries are nonnegative, so $r|T| \leq r|N(T)|$, i.e. $|T| \leq |N(T)|$.

Step 3 (one permutation matrix). By Theorem 11.13, X can be matched into Y; as $|X| = |Y| = n$, the matching is $x_i y_{\sigma(i)}$ for a permutation σ, and $a_{i\sigma(i)} \geq 1$ for all i. Let P_σ be the permutation matrix with 1's at the positions $(i, \sigma(i))$.

Step 4 (induction). $A - P_\sigma$ has nonnegative integer entries, and all its line sums are $r - 1$. By the induction hypothesis it is a sum of $r - 1$ permutation matrices, so A is a sum of r of them. ■